XT-Luna-Zero

Architecture of a 194M-Parameter Sparse Decoder

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Abstract

XT-Luna-Zero is a decoder-only language model designed for training from scratch on a single GPU. Its 16 blocks combine causal attention with a feed-forward mixture of experts containing one shared expert and four routed experts per block. Top-1 routing yields a reported 194,048,640 stored parameters and 99,676,800 active parameters per token under the project's counting convention. We describe the forward path, account for the sparse parameter budget, and distinguish the reported numerical checks from unresolved training and evaluation work. At the latest reported milestone, the project had no real-text training checkpoint. Language quality, reasoning performance, and effective long-context recall remain unmeasured.

1 Purpose and scope

The model is intended as a small research baseline for English language modeling, scientific knowledge, and later conversational and reasoning fine-tuning. Its current contribution is an engineering design and a parameter accounting. This report uses the project's November 6 implementation reports as primary evidence for XT-Luna-specific details. The live checkout and reported measurements were unavailable for independent reproduction during drafting.

2 Decoder architecture

An input sequence of length T is mapped to a stream x of width $d = 640$. Sixteen decoder blocks preserve that width. In the reported block layout, attention first mixes information across token positions. The feed-forward experts then transform each token representation. Writing A_ℓ for causal attention, $N_{\ell,a}$ and $N_{\ell,e}$ for the two pre-normalizations, and M_ℓ for the mixture of experts, the block is

$$u = x + A_\ell(N_{\ell,a}(x)), \qquad x_{\text{next}} = u + M_\ell(N_{\ell,e}(u)).$$

The design uses RMS normalization [1] and SwiGLU experts [2]. After the final block, the output head maps hidden states to next-token scores.

Component	Reported or specified value
Decoder	16 layers, width 640
Experts per layer	One shared, four routed, top-1 routing
Attention schedule	Global causal attention at zero-based layers 3, 7, 11, and 15, with local causal attention elsewhere, as reported
Proposed head layout	10 query heads, 2 key/value heads, head dimension 64
Tokenizer and output head	32,768 vocabulary entries and tied embeddings in the design handoff

3 Attention and expert routing

3.1 Local and global causal attention

In the reported schedule, global layers can attend to all earlier positions within the current document. Local layers restrict the visible history to a recent window. For query position t and key position j, the intended local mask permits $0 \leq t - j < W$, subject to document isolation in packed batches. The precise configured W requires reconciliation. The handoff lists 1,024 tokens, while the implementation report treats 2,048 as the boundary. Packed-document isolation is a production check, rather than a result implied by the single-sequence attention comparison. The proposed grouped-query head layout follows the GQA design family [3].

3.2 Shared and routed feed-forward experts

Each token reaches the shared expert and the routed expert selected by the router. For a normalized token state z, let $p(z)$ be the four-way softmax over router scores and $e^* = \arg\max_e p_e(z)$. The reported design combines the expert outputs as

$$M(z) = E_{\text{shared}}(z) + p_{e^*}(z) E_{e^*}(z).$$

The selected probability scales the routed contribution. Router probabilities are calculated in FP32. The proposed correction applies the routing weight and returns the resulting contribution in the residual stream's dtype before the residual addition. The architecture follows prior work on shared and routed experts and sparse selection [4, 5]. Each expert is a gated feed-forward network of the form

$$E(z) = W_{\text{down}}[\text{SiLU}(W_{\text{gate}} z) \odot W_{\text{up}} z] \,.$$

The implementation handoff proposes a load-balancing term based on the fraction of tokens assigned to each expert and the mean router probability, together with router z-loss. Integration and coefficients must be confirmed in the actual trainer before presenting them as run settings.

3.3 Parameter accounting

The reported width of an expert's intermediate representation is 1,024. Three 640-by-1,024 projections account for 1,966,080 routed expert weights. Across 16 layers, top-1 routing skips three of the four routed experts per token.

$$16 \times 3 \times 1{,}966{,}080 = 94{,}371{,}840.$$

Quantity	Parameters
Stored model	194,048,640
Routed weights skipped per token	94,371,840
Active per token under the project's convention	99,676,800

The count describes selection for a single token. A batch can activate every routed expert, and the stored weights still contribute to memory requirements. The active count is distinct from measured FLOPs or wall-clock throughput.

4 Reported engineering evidence

Check	Evidence and limits
Attention reference	The project reported zero relative RMS output error against an explicit causal-window reference in FP32 at lengths 1,024, 2,048, 4,096, and 6,144. This check covers the tested output comparisons.
Grouped top-1 dispatch	The v3 prototype reportedly matched FP32 reference outputs and passed gradient checks. The last checked `train.py` still imported `model.py` (v1).
BF16 integration	A subsequent trace found that multiplying BF16 expert activations by FP32 routing weights promoted the contribution to FP32 and caused a dtype mismatch in a later layer. A fix was proposed. A complete production BF16 training step remains unverified.
Data and checkpoints	The only reported checkpoint was trained on a repeated numeric pattern for a smoke test. Later reports describe tokenizer improvements and an incomplete mixed-domain data build. No real-text pilot result was supplied.

5 Pretraining and evaluation protocol

The planned English corpus uses a 60/30/10 training-token allocation for broad English, science, and combined math and code. This allocation describes consumed tokens. The raw data inventory is tracked separately. Freeze source revisions and hold out document families before tokenization and packing. Verify that a resumed run consumes the expected next batch and that all buckets meet the pilot quota after filtering and holdout.

The first language pilot is capped at 0.5B tokens or three GPU-hours, whichever comes first. Record held-out next-token loss by domain, effective tokens per second, peak memory, and the exact checkpoint and model backend. Treat each context length as a distinct evaluation condition. A successful memory-fit test establishes feasibility for that configuration. Measuring recall requires a trained checkpoint and controlled tasks with distant facts, corrections, or multiple supporting passages.

6 Scope of future experiments

The proposed larger flat control, phase-specific expert pools, chained reasoning, Grassmann mixing, rhythm conditioning, multi-token prediction, and a DeepSeek-inspired reading and writing split are future experiments. The baseline described here uses a single decoder stack. Current engineering checks support specific numerical comparisons. Claims about useful language modeling or improvements from architectural changes require held-out results from real-text runs.

References

[1] B. Zhang and R. Sennrich. *Root Mean Square Layer Normalization*. 2019. https://arxiv.org/abs/1910.07467

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